KIRTI SINHA

Software Engineer | AI/ML Engineer | Full-Stack Developer

+91 7011778104 | sinhakirti968@gmail.com | LinkedIn | GitHub

Software Engineer with experience building AI-powered products, distributed backend systems, and LLM-based applications. Experienced in Python, FastAPI, Django REST, React, GraphQL, LangGraph, OpenAI, Kafka, PostgreSQL, MongoDB, and AI search/document intelligence workflows. Worked at Zinnia and Microsoft across insurance, enterprise search, GenAI, and applied ML domains.

KEY HIGHLIGHTS

✓ Software Engineer at Zinnia building production AI systems.	✓ Former Software Engineering Intern at Microsoft.
✓ Research papers presented at ICRET 2023 and ICCCNT 2023.	✓ Strong stack: LLMs, LangGraph, FastAPI, React, Kafka, PostgreSQL, MongoDB.

PROFESSIONAL EXPERIENCE

Software Engineer I | Zinnia

Aug 2025 - Jul 2026

- Engineered ZIVA, an AI-powered insurance discovery platform leveraging OpenAI, LangGraph, Django REST, React, Twilio, and MongoDB to support customer onboarding through voice and text interactions.
- Built backend pipelines for structured profile extraction, Whisper-based call transcript analysis, MongoDB-backed transcript search, and AI-ranked Top-3 recommendations.
- Developed EVA, an asynchronous AI-driven insurance-form validation service that converts PDFs into high-resolution page images and verifies mandatory owner, SSN, EFT, and signature fields against configurable business rules.
- Designed shared FastAPI and Kafka processing flows with correlation-ID tracking, PostgreSQL job persistence, MongoDB result retrieval, Kafka success/failure events, and Langfuse prompt versioning and tracing.

Software Engineering Intern | Microsoft

May 2024 - Jul 2024

- Added an Images tab to the Search Results page using React and GraphQL, making copied and uploaded images searchable in a person-centric scope.
- Contributed to enterprise search experience improvements with emphasis on discoverability, UI integration, and GraphQL-backed data retrieval.

Research Intern | Prodigal AI

Mar 2023 - Jul 2023

- Built machine learning prototypes for legal and healthcare use cases using LLMs, Hugging Face, LSTMs, Transformers, and BERT.
- Supported model experimentation and comparative research for domain-specific AI applications.

SELECTED PROJECTS

PDF Insights | *LangChain, OpenAI, Streamlit* | Project | Paper

- Developed an AI engine for answering questions based on PDF and CSV files using LangChain, OpenAI, and Streamlit.
- Presented the associated research paper at ICRET 2023.

LinkedIn Personalized Post Recommender System | *NLP, Deep Learning, BERT, GPT-2* | Project | Paper

- Developed an NLP-based recommender system to suggest relevant LinkedIn posts based on user activity feed.
- Published comparative research evaluating BERT, GPT-2, and traditional machine learning models at ICCCNT 2023.

TECHNICAL SKILLS

Languages: Python, C++, SQL

Backend & APIs: Django REST Framework, FastAPI, REST APIs, Kafka, GraphQL

AI/ML & GenAI: OpenAI, Whisper, LangGraph, LangChain, Hugging Face, BERT, Transformers, LLM Applications

Frontend: React, JavaScript

Databases & Tools: PostgreSQL, MongoDB, Git, GitHub, Langfuse

EDUCATION

B.Tech in Computer Science | Indira Gandhi Delhi Technical University for Women — **2025**

- CGPA: 8.92

CBSE Class XII | Delhi Public School, R.K. Puram — **2019**

- Aggregate: 92%

ACHIEVEMENTS

- Zinnovate 2025: Secured 2nd place at Zinnia innovation challenge.
- Smart India Hackathon 2023: Grand Finalist, Team VividVisions ranked among top 1,000 teams.
- Innerve Ideathon 2022: Secured 1st place at Innerve Prefest, IGDTUW, as part of Carbon Warriors.
- Ideate2Sustain 2022: Secured 2nd place at CSD & Cognito, IGDTUW, as part of Nature's Wizards.
- Erudite Ideathon 2022: Secured 4th place; idea approved for patent processing by ARC Data Science Club, IGDTUW.
- Hack The Mountain 2022: Selected for MLH offline hackathon from 150 teams.
- Flipkart GRID 2023: Qualified the first challenge round.
- Lean In Hacks 4.0 2023: Selected among top 10 teams in Round 1.

LEADERSHIP & CAMPUS INVOLVEMENT

- Content Team Head, OPTICA, IGDTUW.
- Treasurer, Microsoft Learn Student Ambassador, IGDTUW.
- Training & Placement Coordinator; managed student databases and coordinated industry interactions.